

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Experiments

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	<i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)</i>		
Student Name:	M.MANASA	Reg. No.:	192425189
Section / Batch:	2024 AIML	Date:	11/8/2026

Code of Conduct

I M.MANASA(192425189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: m.manasa

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts	
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation	
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools	
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results	
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation	

Edge Detection under Illumination Changes

Design an experiment to study the effect of illumination variation on edge detection performance. Explain the relationship between lighting conditions and edge extraction accuracy. Apply edge detection methods on images with varying illumination levels, document the outputs clearly, and analyze the robustness of the techniques under different lighting conditions.

Aim:

To study the effect of different illumination levels on edge detection performance and analyze how lighting conditions affect the accuracy and robustness of edge extraction techniques.

Algorithm:

1. Read the input image.
2. Create different illumination conditions such as low, normal, and high brightness.
3. Preprocess the images by converting them to grayscale.
4. Apply edge detection techniques, such as:
 - o Canny Edge Detection
 - o Sobel Edge Detection
5. Compare the detected edges under different lighting conditions.
6. Observe changes in edge clarity, missing edges, and noise.
7. Analyze the robustness of each edge detection method.
8. Document the results and determine which method performs better under illumination changes.

CODE:

```
import cv2

path = r"C:\Users\mamid\Downloads\flower.jpg"

img = cv2.imread(path)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

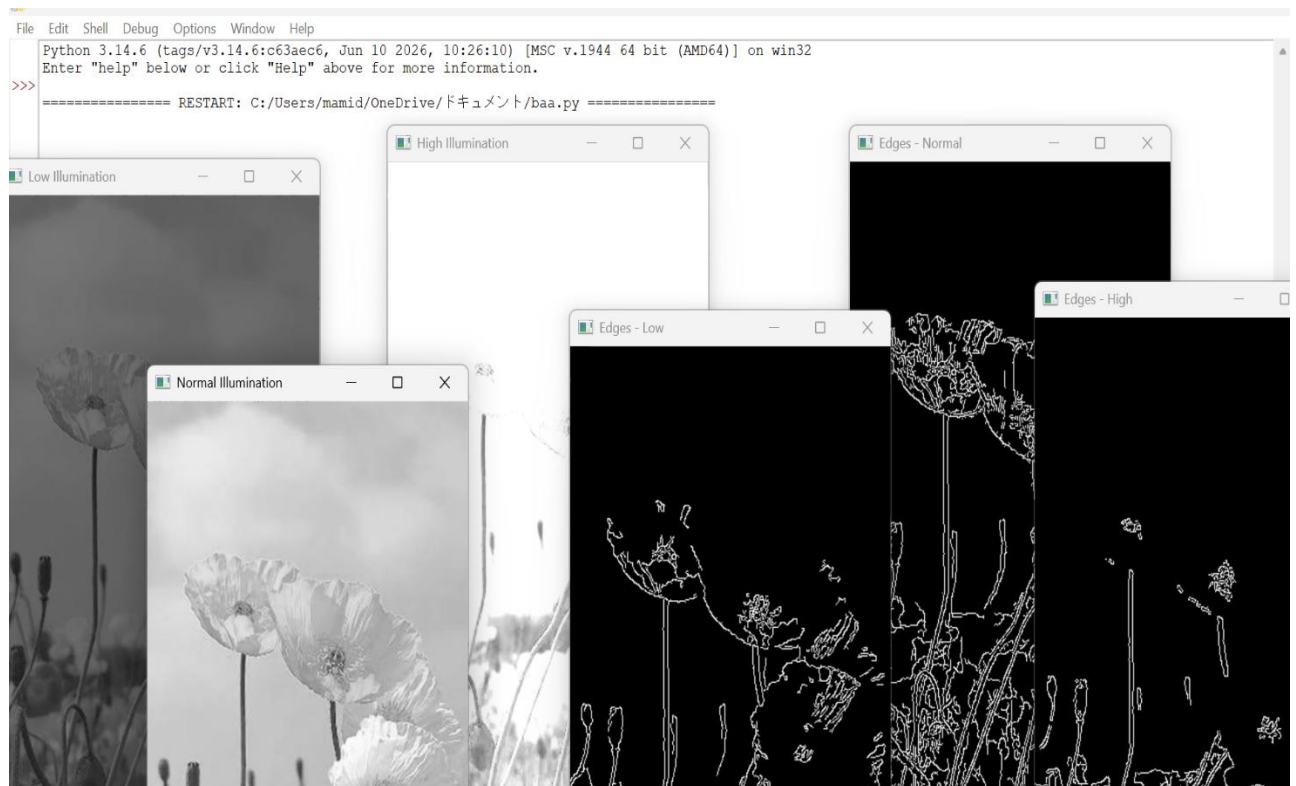
# Illumination changes
low = cv2.convertScaleAbs(gray, alpha=0.5, beta=0)
normal = gray
high = cv2.convertScaleAbs(gray, alpha=1.5, beta=30)

# Canny edge detection
edge_low = cv2.Canny(low, 50, 150)
edge_normal = cv2.Canny(normal, 50, 150)
edge_high = cv2.Canny(high, 50, 150)

# Display
cv2.imshow("Low Illumination", low)
cv2.imshow("Normal Illumination", normal)
cv2.imshow("High Illumination", high)
```

```
cv2.imshow("Edges - Low", edge_low)
cv2.imshow("Edges - Normal", edge_normal)
cv2.imshow("Edges - High", edge_high)
```

```
cv2.waitKey(0)
cv2.destroyAllWindows()
```



LOW ILLUSION:

```
import cv2

path = r"C:\Users\mamid\Downloads\flowerr.jpg"
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cv2.imshow("Edges - High", edge_high)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



NORMAL ILLUSION:

```
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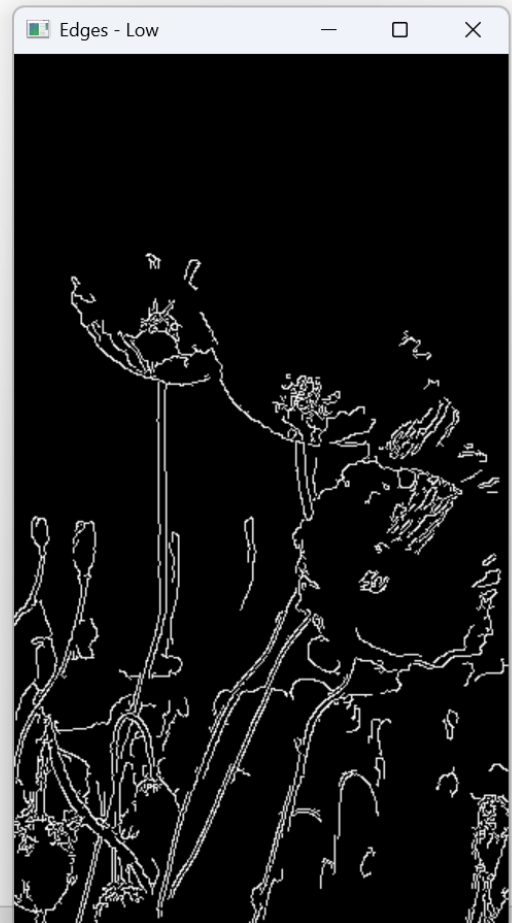
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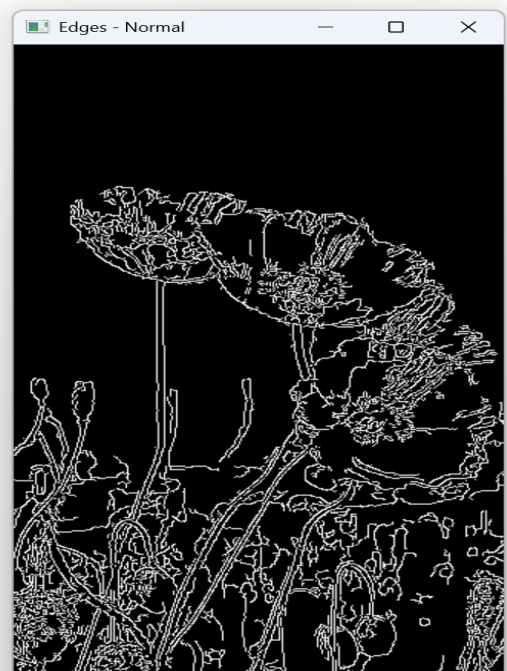
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